

Lecture 10.1: Minimization of DFA

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We can further minimize a DFA by decreasing the number of states. For this we are going to follow Hopcroft's Algorithm. For that, some concepts need to be explained first:

- Sufficiently Different States**

Merge states, if at all possible. Two states cannot be merged if they are “sufficiently different”

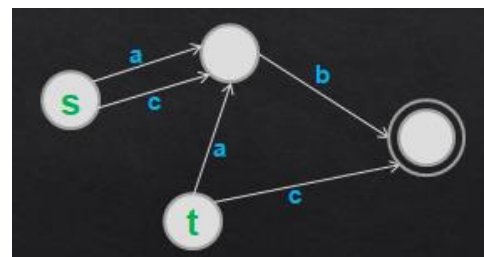
State **s** is “distinguished” from state **t** by some string **w** iff:

Starting at **s**, given characters in **w**, the DFA ends up accepting, but starting at **t**, the DFA does not accept. For example,

“ab” does not distinguish **s** and **t**.

But “c” distinguishes **s** and **t**.

Therefore, **s** and **t** cannot be merged. Because they are sufficiently different states.



- Partitioning**

A partitioning of a set is simply breaking the set into non-overlapping subsets or “groups” based on the “sufficiently different” concept.

Example:

$$S = \{ A, B, C, D, E, F, G \}$$

$$\Pi_1 = \{ (A B) (C D E F) (G) \}$$

$$\Pi_2 = \{ (A B) (C) (D E F) (G) \}$$

- Refine a Partitioning**

$$\Pi_i = \{ \underline{(A B D)} \quad \underline{(C E)} \}$$

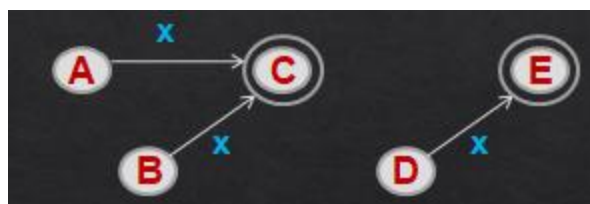
G₁

G₂

Consider one group (A B D)

Look at output edges on some symbol (e.g. “x”)

On “x”, all states in **G₁** go to states belonging to the same group.

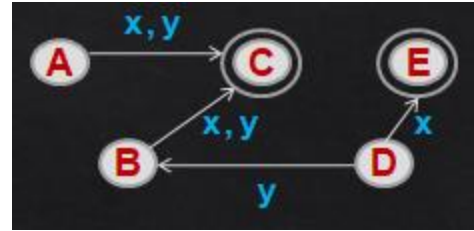


Now consider another symbol (e.g. “y”)

D is distinguished from A and B!

So *refine* the partition!

$\Pi_{i+1} = \{ \underbrace{(A \ B)}_{G_3} \quad \underbrace{(D)}_{G_4} \quad \underbrace{(C \ E)}_{G_2} \}$



Hopcroft's Algorithm

Consider the set of states.

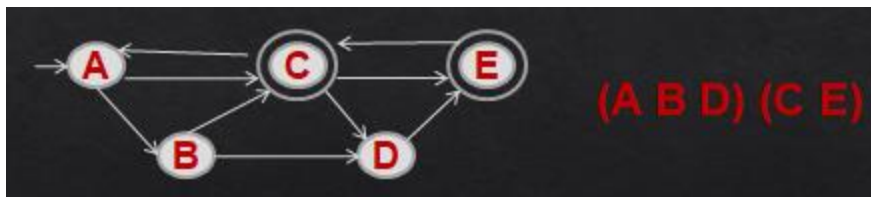
Partition it...

- Final States
- All Other States

Repeatedly “refine” the partitioning.

Two states will be placed in different “groups”

...If they can be “distinguished”



Repeat until no group contains states that can be distinguished.

Each group in the partitioning becomes one state in a newly constructed DFA

DFA_{MIN} = The minimal DFA